

ER Diagram

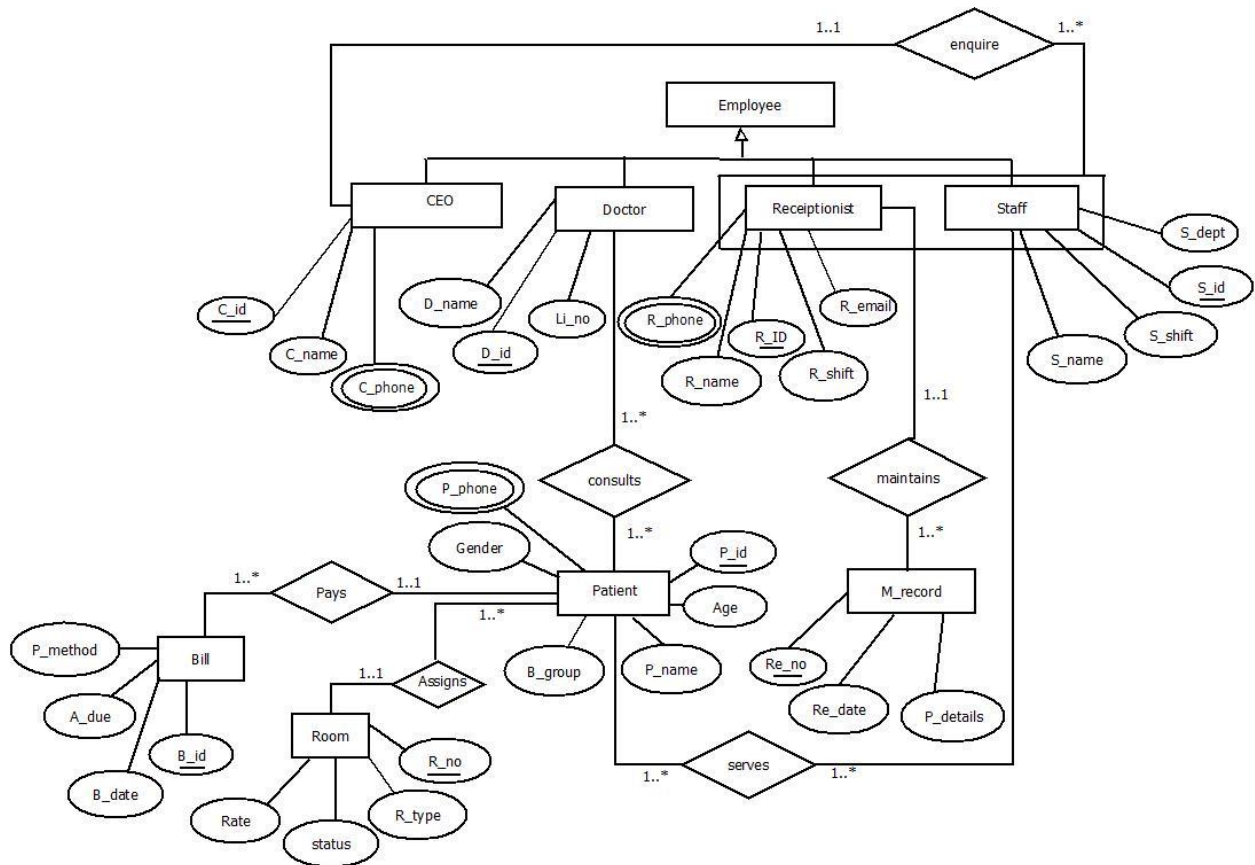


Figure-01: ER Diagram of Hospital Management system

Normalization

Maintains:

UNF: R_Id, R_name, R_shift, R_phone, R_email ,Re_no, Re_date, P_details

1NF: R_Id, R_name, R_shift, R_phone, R_email ,Re_no, Re_date, P_details

2NF:

- 1) R_Id, R_name, R_shift, R_phone, R_email
- 2) Re_no (PK), Re_date, P_details, R_Id (FK)

3NF:

3NF will be the same as 2NF.

Serves:

UNF: S_Id, S_name, S_dept, S_shift, P_Id, P_name, Age, Gender, P_phone, Blood_Group

1NF: S_Id, S_name, S_dept, S_shift, P_Id, P_name, Age, Gender, P_phone, Blood_Group

2NF:

- 1) S_Id, S_name, S_dept, S_shift
- 2) P_Id, P_name, Age, Gender, P_phone, Blood_Group
- 3) P_Id (PK), S_Id (FK)

3NF:

3NF will be the same as 2NF.

Consults:

UNF: D_Id, D_name, Li_no, P_Id, P_name, Age, Gender, P_phone, Blood_Group

1NF: D_Id, D_name, Li_no, P_Id, P_name, Age, Gender, P_phone, Blood_Group

2NF:

- 1) D_Id, D_name, Li_no
- 2) P_Id, P_name, Age, Gender, P_phone, Blood_Group
- 3) D_Id (Pk), P_Id (Fk)

3NF:

3NF will be the same as 2NF.

Assigns:

UNF: P_Id, P_name, Age, Gender, P_phone, Blood_Group, R_no, R_type, Rate, Status

1NF: P_Id, P_name, Age, Gender, P_phone, Blood_Group, R_no, R_type, Rate, Status

2NF:

- 1) P_Id (PK), P_name, Age, Gender, P_phone, Blood_Group, R_no (FK)
- 2) R_no, R_type, Rate, Status

3NF:

3NF will be the same as 2NF.

Pays:

UNF: B_Id,A_Due,B_date,P_method, P_Id, P_name,Age,Gender,P_phone,Blood_Group

1NF: B_Id,A_Due,B_date,P_method, P_Id, P_name,Age,Gender,P_phone,Blood_Group

2NF:

- 1) B_Id(PK),A_Due,B_date,P_method,P_Id (FK)
- 2) P_Id, P_name,Age,Gender,P_phone,Blood_Group

3NF:

3NF will be the same as 2NF.

Enquires:

UNF:C_Id,C_name,C_phone, R_Id, R_name, R_shift, R_phone, R_email,S_Id,
S_name,S_dept,S_shift

1NF:C_Id,C_name,C_phone, R_Id, R_name, R_shift, R_phone, R_email,S_Id,
S_name,S_dept,S_shift

2NF:

- 1) C_Id,C_name,C_phone
- 2) R_Id(PK), R_name, R_shift, R_phone, R_email,S_Id (FK),
S_name,S_dept,S_shift,C_Id(FK)

3NF:

3NF will be the same as 2N

Finalization

- 1) R_Id, R_name, R_shift, R_phone, R_email
- 2) Re_no (PK), Re_date, P_details, R_Id (FK)
- 3) S_Id, S_name, S_dept, S_shift
- 4) P_Id, P_name, Age, Gender, P_phone, Blood_Group
- 5) P_Id (PK), S_Id (FK)
- 6) D_Id, D_name, Li_no
- ~~7) P_Id, P_name, Age, Gender, P_phone, Blood_Group~~
- 8) D_Id (Pk), P_Id (Fk)
- 9) P_Id (PK), P_name, Age, Gender, P_phone, Blood_Group, R_no (FK)
- 10) R_no, R_type, Rate, Status
- 11) B_Id (PK), A_Due, B_date, P_method, P_Id (FK)
- ~~12) P_Id, P_name, Age, Gender, P_phone, Blood_Group~~
- 13) C_Id, C_name, C_phone
- 14) R_Id (PK), R_name, R_shift, R_phone, R_email, S_Id (FK),
S_name, S_dept, S_shift, C_Id (FK)

Final Table:

- 1) R_Id, R_name, R_shift, R_phone, R_email (**Receptionist**)
- 2) Re_no (PK), Re_date, P_details, R_Id (FK) (**Maintain**)
- 3) S_Id, S_name, S_dept, S_shift (**Staff**)
- 4) P_Id, P_name, Age, Gender, P_phone, Blood_Group (**Patient**)
- 5) P_Id (PK), S_Id (FK) (**Serve**)
- 6) D_Id, D_name, Li_no (**Doctor**)
- 7) D_Id (Pk), P_Id (Fk) (**Consultation**)
- 8) P_Id (PK), P_name, Age, Gender, P_phone, Blood_Group, R_no (FK) (**Assign**)
- 9) R_no, R_type, Rate, Status (**Room**)
- 10) B_Id (PK), A_Due, B_date, P_method, P_Id (FK) (**Payment**)
- 11) C_Id, C_name, C_phone (**CEO**)
- 12) R_Id (PK), R_name, R_shift, R_phone, R_email, S_Id (FK),
S_name, S_dept, S_shift, C_Id (FK) (**Enquire**)